

4.sql-stored-procedure

Create a Stored Procedure

```
CREATE TABLE Employees (  
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,  
    FirstName VARCHAR(50),  
    LastName VARCHAR(50),  
    DepartmentID INT,  
    Salary DECIMAL(10,2),  
    JoinDate DATE  
);
```

```
DELIMITER $$
```

#Insert Employee Stored Procedure

```
CREATE PROCEDURE sp_InsertEmployee(  
    IN p_FirstName VARCHAR(50),  
    IN p_LastName VARCHAR(50),  
    IN p_DepartmentID INT,  
    IN p_Salary DECIMAL(10,2),  
    IN p_JoinDate DATE  
)
```

```
BEGIN
```

```
    INSERT INTO Employees
```

```
    (  
        FirstName,  
        LastName,  
        DepartmentID,  
        Salary,  
        JoinDate
```

```
    )
```

```
VALUES
```

```
    (  
        
```

```
        p_FirstName,  
        p_LastName,  
        p_DepartmentID,  
        p_Salary,  
        p_JoinDate  
    );  
END $$
```

```
DELIMITER ;
```

```
#Execute Insert Procedure
```

```
CALL sp_InsertEmployee(  
    'Bhargav',  
    'CH',  
    1,  
    50000,  
    '2026-06-15'  
);
```

```
DELIMITER $$
```

```
#Retrieve Employee By Department
```

```
CREATE PROCEDURE sp_GetEmployeesByDepartment(  
    IN p_DepartmentID INT  
)  
BEGIN  
    SELECT *  
    FROM Employees  
    WHERE DepartmentID = p_DepartmentID;  
END $$
```

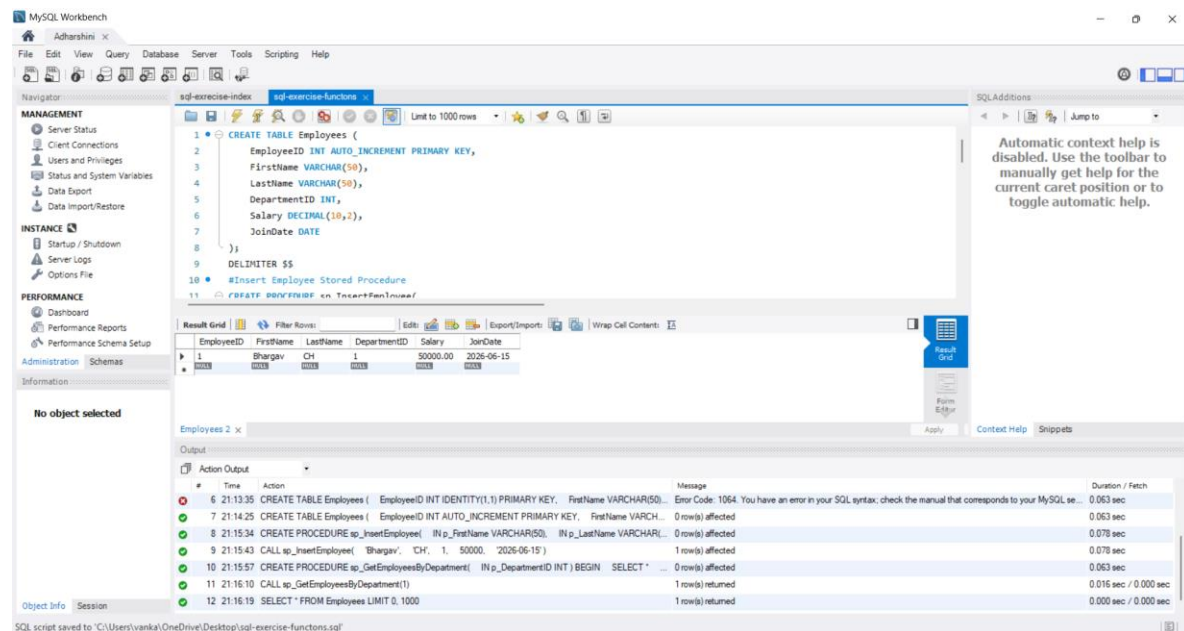
```
DELIMITER ;
```

```
#Execute Retrieval Procedure
```

CALL sp_GetEmployeesByDepartment(1);

#Verify Data

SELECT * FROM Employees;



#exrecise-07-return data from scalar function

CREATE FUNCTION fn_GetSalary(
p_EmployeeID INT
)
RETURNS DECIMAL(10,2)
DETERMINISTIC
BEGIN
DECLARE empSalary DECIMAL(10,2);

SELECT Salary
INTO empSalary
FROM Employees
WHERE EmployeeID = p_EmployeeID;

RETURN empSalary;

END \$\$

DELIMITER ;

SELECT fn_GetSalary(1) AS Salary;

The screenshot displays the MySQL Workbench interface. The main editor window shows a SQL script with the following content:

```
77 INTO empSalary
78 FROM Employees
79 WHERE EmployeeID = p_EmployeeID;
80
81 RETURN empSalary;
82 END $$
83
84 DELIMITER ;
85
86 SELECT fn_GetSalary(1) AS Salary;
```

The Results window shows the output of the query:

Salary
50000.00

The Output window shows the execution log:

#	Time	Action	Message	Duration / Fetch
12	21:16:19	SELECT * FROM Employees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
13	21:22:55	CREATE PROCEDURE sp_CountEmployeesByDepartment(IN p_DeptmentID INT) BEGIN SELECT C...	0 row(s) affected	0.015 sec
14	21:23:11	CALL sp_CountEmployeesByDepartment(1)	1 row(s) returned	0.063 sec / 0.000 sec
15	21:23:22	INSERT INTO Employees (FirstName, LastName, DepartmentID, Salary, JoinDate) VALUES ('Bhargav', 'CH', 1...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
16	21:23:36	SELECT COUNT(*) FROM Employees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
17	21:26:07	CREATE FUNCTION fn_GetSalary(p_EmployeeID INT) RETURNS DECIMAL(10,2) DETERMINISTIC BE...	0 row(s) affected	0.015 sec
18	21:26:19	SELECT fn_GetSalary(1) AS Salary LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The status bar at the bottom indicates "Query Completed".